PineApplee Billing & Logic Specification: Multi-Agent Token System

1. Overview

This document outlines the technical logic for the PineApplee dashboard billing engine. The system is designed as a "Consumption Equalizer" using a Multiplier Engine. This allows us to offer both low-cost and high-end AI models through a single, unified user token balance while maintaining consistent profit margins.

2. The Multiplier Logic (Simplified)

The billing engine does not deduct 1 token for every 1 API token used. Instead, it applies a **Multiplier** based on the cost of the model being utilized. Think of the user's token balance as a "Fuel Tank." Different engines (models) burn that fuel at different rates.

- **Base Speed (1x):** Using a model like DeepSeek or GPT-5 Mini burns the fuel at the normal rate. If a user has 100 tokens and uses a 1x model, the balance drops by exactly the amount used.
- **High Speed (60x):** Using a premium model like Claude Opus burns fuel 60 times faster. If a user uses 1 token of Opus, the system deducts 60 tokens from their balance.

3. Technical Model Mapping & Multipliers

Developers must implement the following multiplier table in the backend logic. When the OpenRouter API returns the total_tokens usage, the final deduction from the database must be: (total_tokens * multiplier).

Model Brand	Model Name	Multiplier	Logic Assignment
DeepSeek	V3.2 / V4 / R1	**1x**	Base Logic (Coding/General)
OpenAI	GPT-5 Mini / 4o-mini	**1x**	Base Logic (Fast Tasks)
Google	Gemini 3.1 Flash	**3x**	Fast Reasoning / Data Parsing
Google	Gemini 3.1 Pro	**12x**	Long Context / Repository Analysis
Anthropic	Claude 3.5 Sonnet / 4.6	**15x**	Advanced Coding (OpenCode)
OpenAI	GPT-5.4 / GPT-4o	**15x**	Advanced Logic / Complex Tasks
Anthropic	Claude 3 / 4.6 Opus	**60x**	Deep Architectural Engineering

Model Provider	Model Name / Version	PineApplee Multiplier	Primary Use Case
Moonshot AI	Kimi (Latest)	1x (Base)	Long-Form Content & Deep Research
DeepSeek	DeepSeek V3.2 / V4 / R1	1x (Base)	Elite Coding & Logic
Alibaba	Qwen 3.5 / 2.5 Coder	1x (Base)	Rapid Scripting & Multilingual Tasks
Zhipu AI	GLM-4 / GLM-5	1x (Base)	High-Speed Logical Reasoning
OpenAI	GPT-5 Mini / 4o-mini	1x (Base)	Routine Automation & Fast Chat
01.AI (Yi)	Yi-Lightning / Large	1x (Base)	High-Performance Budget Tasks
Google	Gemini 3.1 Flash	3x	Fast Data Parsing & Summary
MiniMax	MiniMax M2.5	4x	Specialized Creative & Text Ops
Mistral AI	Mistral Large 2 / 3	10x	Concise Logic & Precision
Google	Gemini 3.1 Pro	12x	Massive Context (2M+) Analysis
Anthropic	Claude 3.5 Sonnet / 4.6	15x	Advanced Engineering (OpenCode)
OpenAI	GPT-5.4 / GPT-4o	15x	Multi-Step Reasoning & Orchestration
Meta	Llama 3.1 / 4.0 (405B)	15x	High-End Open-Source Logic
Anthropic	Claude 3 / 4.6 Opus	60x	Deep Architectural Engineering

(This is the detailed table for reference, use this for the models)

Link:
https://docs.google.com/spreadsheets/d/1X9TrBAhW1yZi6CMrPAm9ZRS880F-dgRNnNCTFOS3huM/edit?usp=sharing

4. Subscription Tier Constraints

The backend must restrict model access based on the user's active subscription tier to prevent "Multiplier Overrun."

- **Free (free overall):** Access to one default agent + monthly or daily specific tokens (we need to discuss it with dev for this, as this would also be editable from the admin panel so that we can change the things)
- **Starter (₹799):** Access up to 4 agents from marketplace
- **Silver (₹1,999):** Access up to 7 agents from marketplace
- **Pro (₹4,999):** Access up to 20 agents from marketplace
- **Business (custom):** Contact Sales (land to our email)

5. Critical Implementation Requirements

1. **Database Locks:** When deducting tokens, use a database transaction to prevent "double-spending" if a user sends multiple concurrent requests.
2. **Prompt Caching:** Developers MUST enable OpenRouter prompt caching. This reduces input costs by 90% for repeated code contexts.
3. **Intent Check:** Before calling the API, check if the requested task matches the allowed multipliers for that user's tier.

Marketplace:
The user needs to select the subscription, then based on the subscription the user can see how many agents they can choose, for basic one they can choose any 3-4 agents, so they can go to the agents section and select the agents and choose that and make the final payment and done. (workflow already there in the previous doc).

Subscription Structure & Key Features

Tiers are designed to provide access to the Agent Marketplace while managing infrastructure costs. This needs to be editable from the admin panel for easy smooth working.

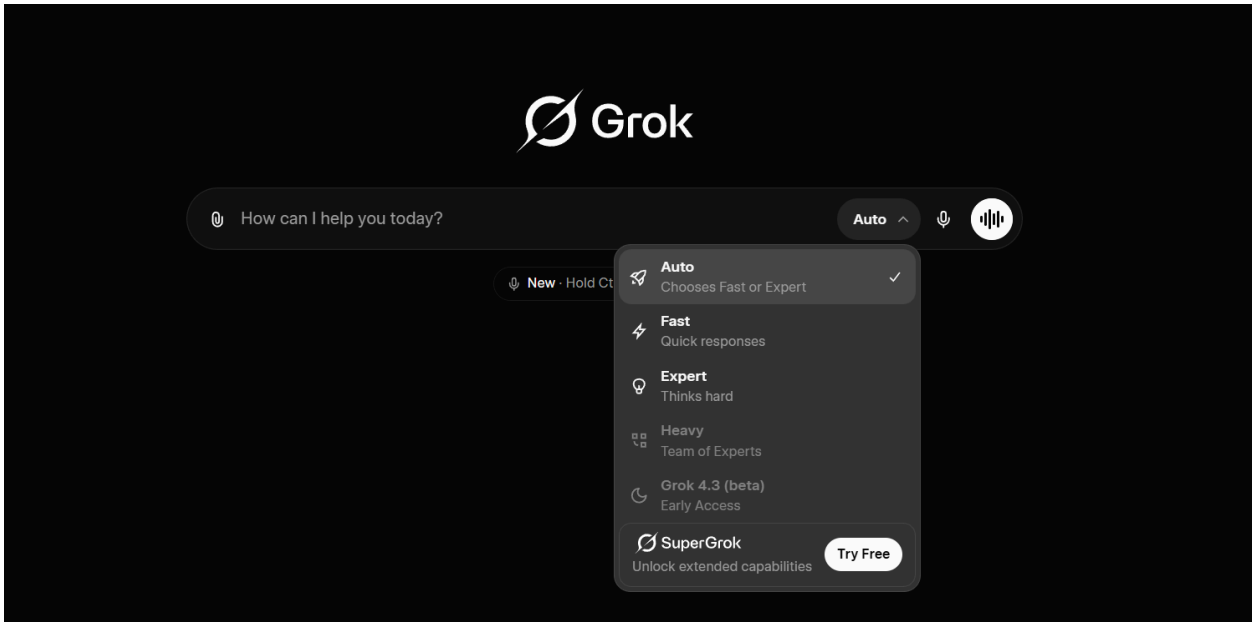
Tier	Price (Monthly)	Token Bucket	Access Features
Starter	**₹799**	600,000	4 Agent
Silver	**₹1,999**	2.5 Million	7 Agent
Pro	**₹4,999**	6 Million	20 Agent

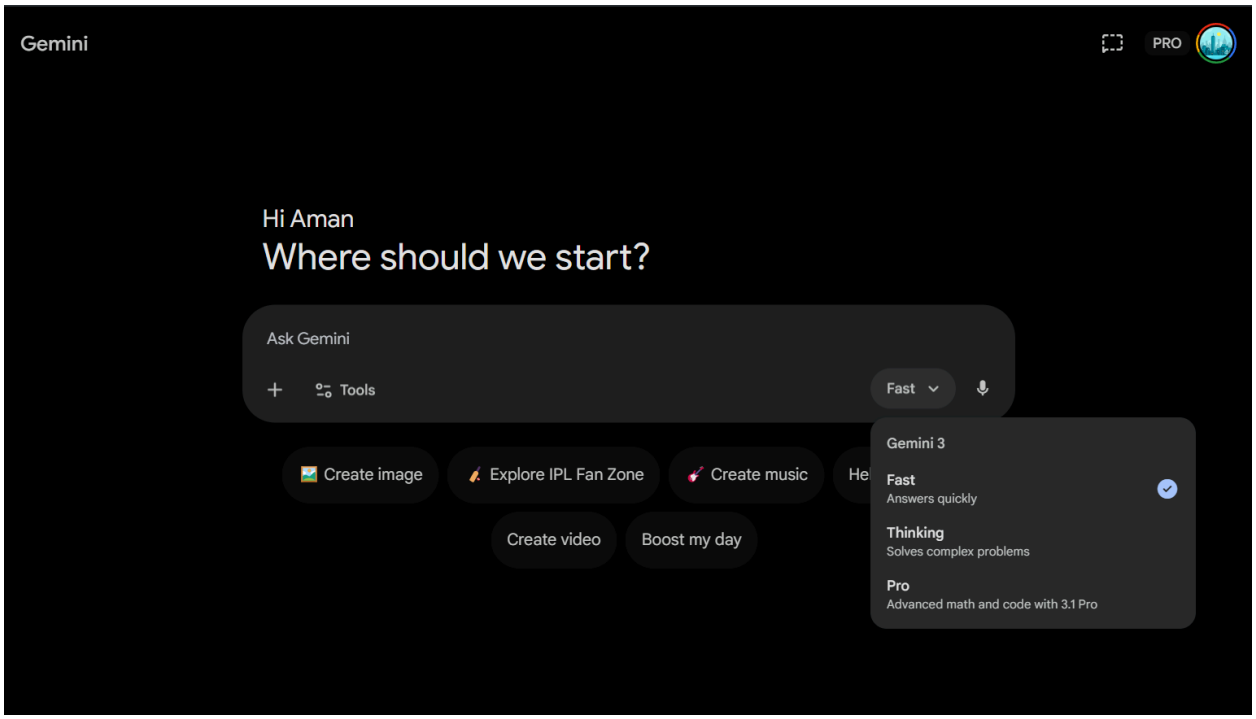
We would also need to set API call limits, as the tokens are there, and if it exhausts then the user can easily buy it, but we also need to work on API hit rate limits, make it like per day per user only 50 limits or flexible so that the user who is using heavily will be shown “You’ve hit the rate limit, please try again after 2 hrs, or rate limit will reset on xyz am/pm” just like we see in Claud, or Google Antigravity.

We need to work on Queuing (already mentioned in the doc) so that the user won't be blindsided about what's going on, so if the system is under heavy usage the user will see that your tasks are in queue due to heavy usage. At least we should provide them some notification so that it's clear that it's not an error but a queue.

Proper token consumption should be shown for particular tasks below the output or response.

We are planning to implement a feature in the input box where the user can choose between the agents, so that it would save our auto switching costing and efforts, so its like the same we see in cursor ai and Grok or Gemini where we can easily choose the model that we want to use so that it won't consume more tokens. As we use OpenCode, I think this would be useful as all the AI models are good in coding so what logic we would be using for auto switching, so let the user choose the models.





Some images of Grok, Gemini for reference.

Please make sure that the UI UX would be very smooth and creative with dark and light mode, and also work best on UI UX as its all about the user experience that they would be paying for, so make a look on that, and also the file preview, the files and things are already mentioned in the main doc. (Refer Page 4 and 5), as we are using OpenCode so all these things need to be taken care of very thoroughly.

New Token Recharge Packs (Growth Pricing)

These packs are designed for users who hit their limits before the month ends. We have reduced these prices to be "disruptive" in the market. So when the user runs out of tokens while using the agent, although for low token balance the user will get notification or error message, then they can buy new tokens and top up their pack and use the agents again. These token pricing should be editable from the admin panel.

Pack Name	Price (INR)	Tokens Added	Effective Rate (per 1k)
Lite	**₹99**	100,000	₹0.99
Speed	**₹249**	300,000	₹0.83
Power	**₹749**	1,000,000	**₹0.75**
Giga	**₹2,499**	4,000,000	**₹0.62**